F70LA Life Insurance Mathematics A (2025/26)
Tutorial 1

1. Let the random variable $T_0 \geq 0$ represent the future lifetime of an newly-born child. The distribution function of T_0 is

$$F_0(t) = \begin{cases} \frac{t}{80} & \text{for } 0 \leq t \leq 80 \\ 1 & \text{for } t > 80. \end{cases}$$

   (a) Calculate the probability that the newborn will:
      (i) Die not later than exact age 65,
      (ii) Survive to at least exact age 50,
      (iii) Die within five years from exact age 40,
      (iv) Die in the 34th year of life.

   (b) Determine the probability density function of T_0 and comment on the result.

   (c) Calculate the expected value of T_0 and interpret the result.

   (d) Calculate the standard deviation of T_0.

2. Suppose that the cumulative distribution function of the future lifetime of a newborn is

$$F_0(t) = 1 - \exp\left(-(0.002t + 0.000023t^2)\right), \qquad \text{for } t \geq 0.$$

   Calculate the probability that a newly born individual will:

   (a) Die before exact age 60,

   (b) Survive to at least age 80,

   (c) Die between exact ages 55 and 65,

   (d) Die either between exact ages 40 and 45 or between exact ages 50 and 55.

3. Suppose that the cumulative distribution function of the future lifetime random variable T_0 is $F_0(t) = 1 - e^{-\lambda t}$, where $\lambda > 0$ is a constant, for $t \geq 0$.

   (a) Write down an expression for $F_0(60)$ and state what $F_0(60)$ represents.

   (b) What happens to the value of $F_0(60)$ if the constant λ is allowed to increase?

   (c) Derive an expression for the probability density function of T_0.

   (d) Show that the force of mortality at age x is λ.

4. Suppose that $F_0(t) = 1 - \exp(-0.01t)$, for $t \geq 0$.

   (a) Derive expressions for $F_x(t)$ and $S_x(t)$.

   (b) Comment on the expressions.

   (c) Calculate the probability that:

      (i) A life currently aged 30 will die before exact age 60,

      (ii) A life currently aged 50 will survive to at least age 80,

      (iii) A life currently aged 40 will die between exact ages 55 and 65,

      (iv) A life currently aged 30 will die either between exact ages 40 and 45 or between exact ages 50 and 55.

5. T_x is the random variable representing the complete future lifetime for a life currently aged $x \geq 0$. You are given that the cumulative distribution function of T_0 is

$$F_0(t) = 1 - e^{-(\alpha t + \beta t^2)},$$

   where α and β are constants. Show that the cumulative distribution function of T_x is

$$F_x(t) = 1 - e^{-\alpha t - \beta(2xt + t^2)}.$$

Numerical solutions: 1(a)(i) 65/80; 1(a)(ii) 30/80; 1(a)(iii) 5/80; 1(a)(iv) 1/80; 1(c) 40 years; 1(d) 23.094 years. Q2(a) 0.18356; 2(b) 0.73550; 2(c) 0.03884; 2(d) 0.03607; Q4(c)(i) 0.25918; Q4(c)(ii) 0.74082; Q4(c)(iii) 0.08191; Q4(c)(iv) 0.08406.